

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8508**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2023601003
Course Section: 1					

L_29_527_986_L_1505_38423



Name of the candidate. Nikhil Hemant Chaudhari

Candidate's UID number:

2	0	2	3	6	0	1	0	0	3
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Examination class room number:

4	0	1
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Invigilator's signature with date:

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(To be filled in by the candidate)

EXAMINATION: BE - DTECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : 7

PROGRAM: CSE- AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 8/10/25 Wed

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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(a+b)
(b)

- Overfitting is a condition in which the model learns all the things from the data given, it's limited to the data that's given.
- This condition typically gives maximum accuracy in the training dataset but struggles really hard if given a data parameter outside the training data.
- So for example if this overfitted model is provided with something different apart from what's being provided then that thing struggles as it doesn't get room to train and self-validate.

The error calculation is basically done using various techniques one being the MSE - Mean Squared Error technique which typically just generalizes the difference between Actual and predicted values.

$$\|x - \hat{x}\|^2$$

=> Overfitting condition can be improved by splitting the data into sets of train-test so that it also gets headroom for the model to properly evaluate.

=> Dimensionality reduction can also be applied suggesting that the model doesn't capture the noise and unwanted error from the dataset.

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Q.2ja

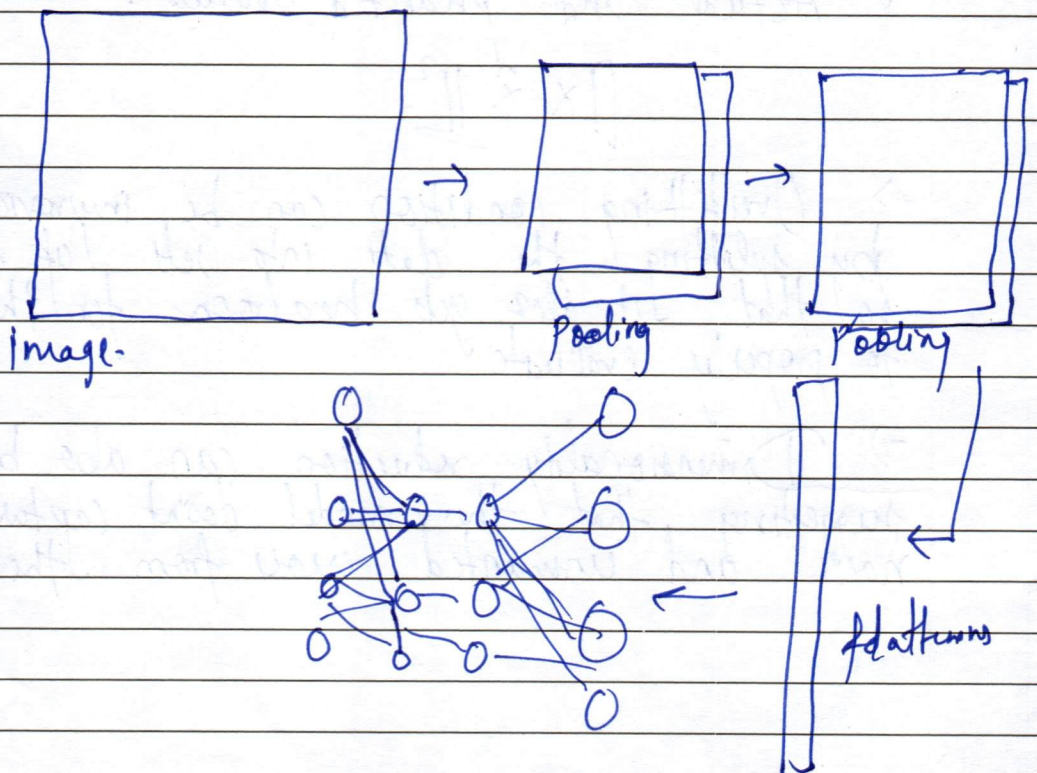
Depthwise separable convolutions is critical because it is the rule to reduce the dimension of any input

for ex :- Suppose we have

1000 x 1000 pixel image
→ this will lead to 1000000 pixel data

If we flatten this up directly and feed to any neural network this would definitely add up to lot of cost therefore it is highly computational heavy to ~~prop~~ directly add it. Pooling layers and filters are therefore necessary in order to properly make it up for it.

Example :-





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Let's say we have an input size
 $H \times W \times C$ with K filters of size $F \times F$.
then the output image size would be

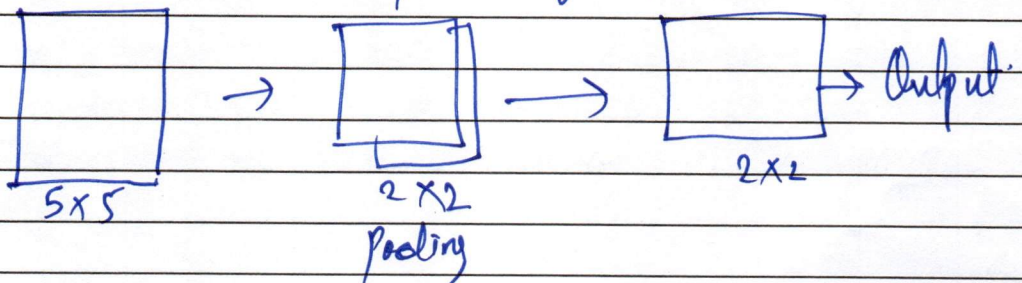
$$[i - F] + 1 \rightarrow \textcircled{1}$$

$H \times W \times C = i$
Size of filter = F .

Multiple operations has to be performed before feeding the input to the neural network because it only helps in properly understanding and grabbing only important and necessary features for us.

Kernels help in reducing the size/dimension so that it's properly fed.

if we have an input image of size 5×5



the output image would be of size 2×2 .
i.e. it would have $\lfloor N/2 \rfloor$ if pooled.

in the same formula ... from $\textcircled{1}$
if we divide it by size of stride

$$\left[\frac{i - F}{s} \right] + 1 \rightarrow \text{this gives } w \rightarrow \textcircled{2}$$

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the size of output image when strides are applied.

If Padding operation is performed

$$\left\lfloor \frac{i - K + 2P}{S} \right\rfloor + 1 \rightarrow \text{from (3)}$$

the Padding size is added into the same formula to get the dimensions of the output image.



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Q.2] b]

- Undercomplete Autoencoder is basically an autoencoder in which the bottleneck size is less than the original input.

$$MSE = ||x - \hat{x}||^2 = \sum (x - \hat{x})^2$$

The mean squared error is basically the function if used with it it learns the subspace equivalent to that of Principal Component Analysis.

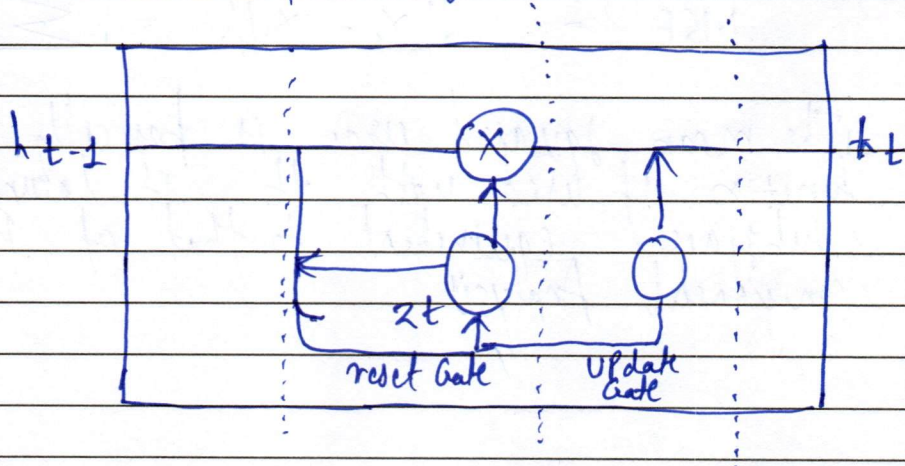
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Q3) b]

→ GRU (Gated Recurrent unit) is basically an advancement in terms of space and optimization when compared to LSTM



→

GRU Architecture

→ If we compare the Architecture of LSTM and GRU the whole cell state is being eliminated in one single state divided into two gates

- Reset Gate
- Update Gate

$$C_t = \left[\left[W_c \cdot [h_t \cdot H_t] + W_b \right] \right]$$

Cell state works on this thing and then the forget gate tells in LSTM properly analyzes what to delete and what to delete the things.

But Here in GRU everything is properly done with just two gates and in a single state

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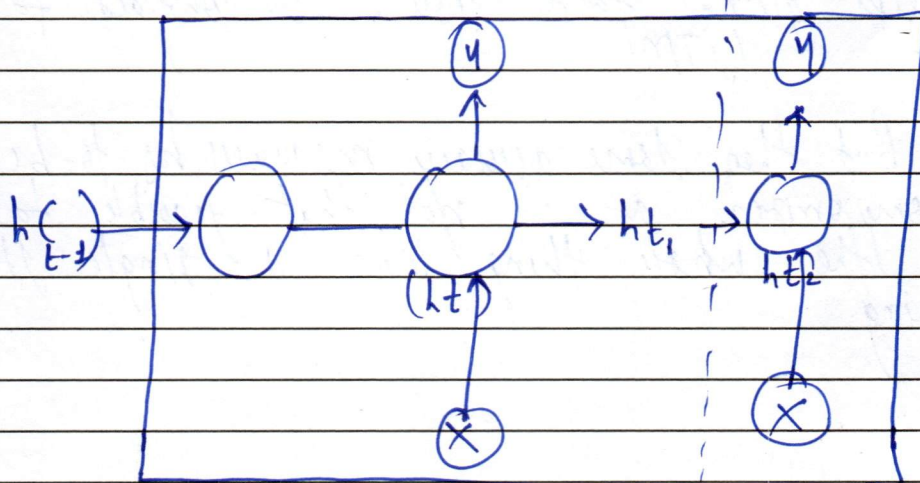
- This is very crucial as it optimizes speed and helps giving faster results as compared to that of LTM.
- But there some accuracy measures has to be also compromised as it's not that feasible to process the whole thing into a single state thing.

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Q.3]a]

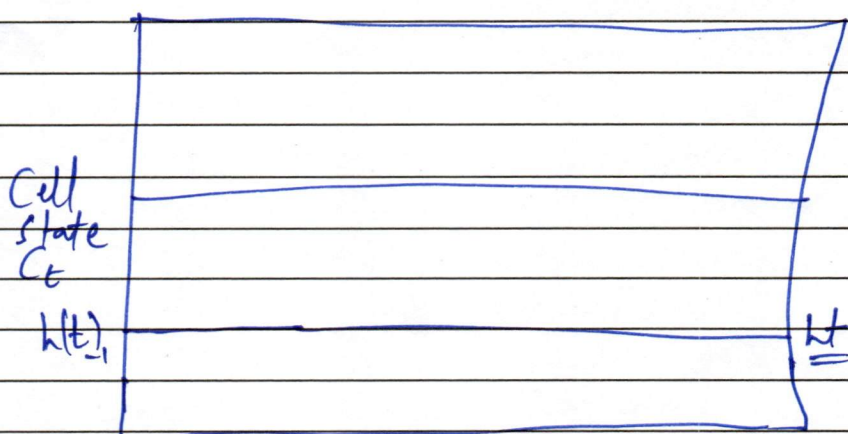
RNN



The major drawback in the RNN is that the previous state cannot be known as only the current module thing is computed and processed further. Hence

WTM introduces a cell state which has three gates

- + Input Gate
- Forget Gate
- Output Gate



with the further computations are made with the help of these gates

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		$\frac{\partial L}{\partial h_{t-k}}$ Recursively computed in RNN
		but due to just an introduction of forget gate in LSTM this gets changed into

[illegible]

[illegible]

[illegible]

[illegible]